PRAKHAR BHATNAGAR

Software Engineer | Python & AI/ML Developer | Full Stack Developer
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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Software Engineer and B.Tech Information Technology student with hands-on experience in Python development, AI/ML, computer vision, and full-stack development. Skilled in building scalable backend services, REST APIs, and automation systems, with proven impact delivering AI-powered industrial safety and security solutions. Strong foundation in Data Structures & Algorithms, OOP, and software engineering, backed by published research in AI-driven optimization. Immediate joiner, seeking a Software Development Engineer (SDE) / AI-ML Engineer role.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C, SQL
Web Development: HTML5, CSS3, JavaScript, React.js, Next.js, Node.js, FastAPI, REST APIs
AI / Machine Learning: Machine Learning, Computer Vision, YOLO, OpenCV, Scikit-learn, TensorFlow, Pandas, NumPy, Random Forest, Data Analysis
CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks
Tools & Technologies: Git, GitHub, Docker, Linux, VS Code, Postman, Microsoft Teams, Figma, Excel, PowerPoint

PROFESSIONAL EXPERIENCE

Automation Intern — ACME Solar MMP, Jaipur *1 Month*

- Developed AI-powered software for industrial automation, including computer vision and gate-security systems, within the Automation Department.
- Built REST APIs and backend services using Python and FastAPI to support AI model deployment, testing, and production rollout.
- Integrated AI models with industrial and gate cameras to enable real-time detection, monitoring, and automated reporting.

PROJECTS

AI-Based Multi-Objective ECU Calibration for Dual-Fuel Engines

- Built an AI-based surrogate modeling framework using Random Forest regression to predict engine performance (BSFC, NOx, Thermal Efficiency), achieving R^2 up to 0.98 on NOx emissions under 5-fold cross-validation.
- Benchmarked model against Gradient Boosting, XGBoost, and Ridge regression; developed a Pareto-front multi-objective optimization module for efficiency-emissions trade-offs.
- Authored findings as an academic paper submitted to IEEE ICTACS 2026 (currently under review).

AI-Driven Adaptive ECU Calibration Framework (Patent Filed)

- Designed a novel adaptive calibration framework extending prior ECU calibration research; patent application filed and currently pending grant.
- Engineered a data-driven pipeline for real-time ECU parameter adaptation across varying fuel compositions and operating conditions.

ACME Guard — Digital Gate Register & Vehicle Access System

- Built a digital gate register processing dual RTSP camera feeds with an automated license plate detection and OCR pipeline for real-time vehicle identification.
- Developed and hosted a full-stack server application giving guards real-time allow/deny decisions with vehicle details, snapshots, and automated daily reports via Microsoft Teams.

ACME Vision AI — Industrial Safety Monitoring System

- Built a real-time computer vision pipeline (YOLO, OpenCV) deployed via FastAPI REST services, integrated with live industrial cameras for automated hazard detection and continuous monitoring.

EDUCATION

Manipal University Jaipur *Expected Graduation: 2027*
B.Tech in Information Technology

L.P. Savani Academy, Surat — Class XII *2023*
Adani Public School, Mundra — Class X *2020*

LEADERSHIP, ACTIVITIES & CERTIFICATIONS

- Vice President & Co-Founder, DevForge Technical Club — established the club and now lead strategy, operations, and cross-functional coordination; previously General Secretary.
- Participant, IEEE Hackerz Street 3.0 Hackathon.
- NPTEL Certifications — Software Engineering; Programming, Data Structures and Algorithms using Python.